

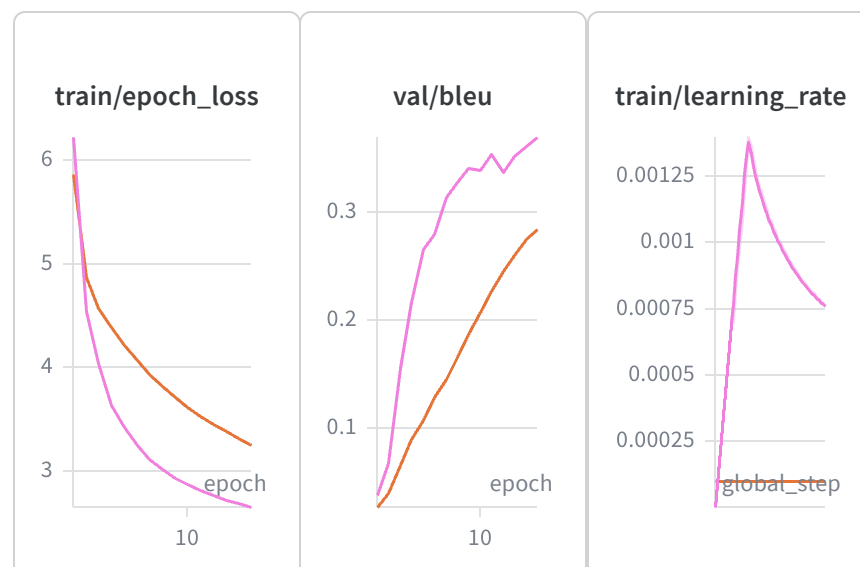
DA6401 Assignment 3 — Transformer for Machine Translation

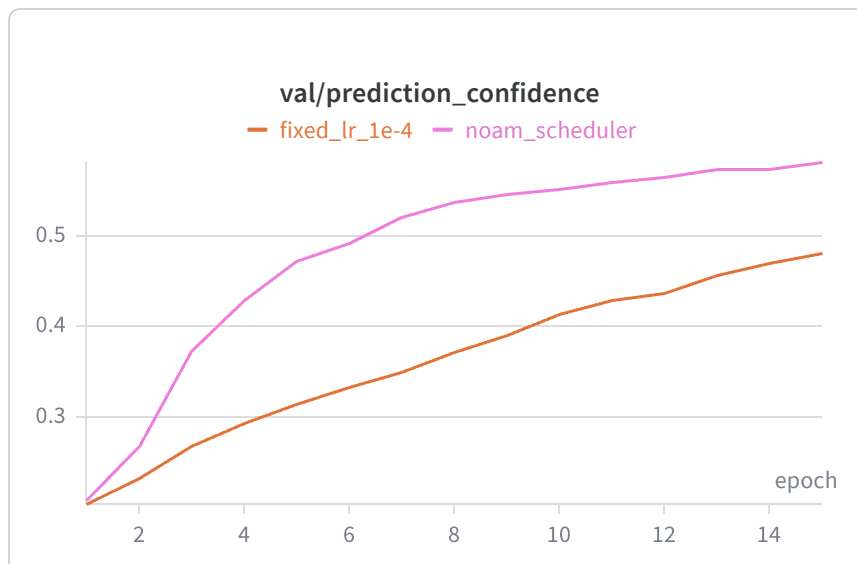
This report presents the implementation and experimental evaluation of a Transformer-based Neural Machine Translation model for German-to-English translation on the Multi30k dataset, following the Attention Is All You Need architecture. The study investigates key architectural and optimization components of the Transformer, including the Noam learning rate scheduler, scaled dot-product attention, positional encoding strategies, attention head behavior, and label smoothing.

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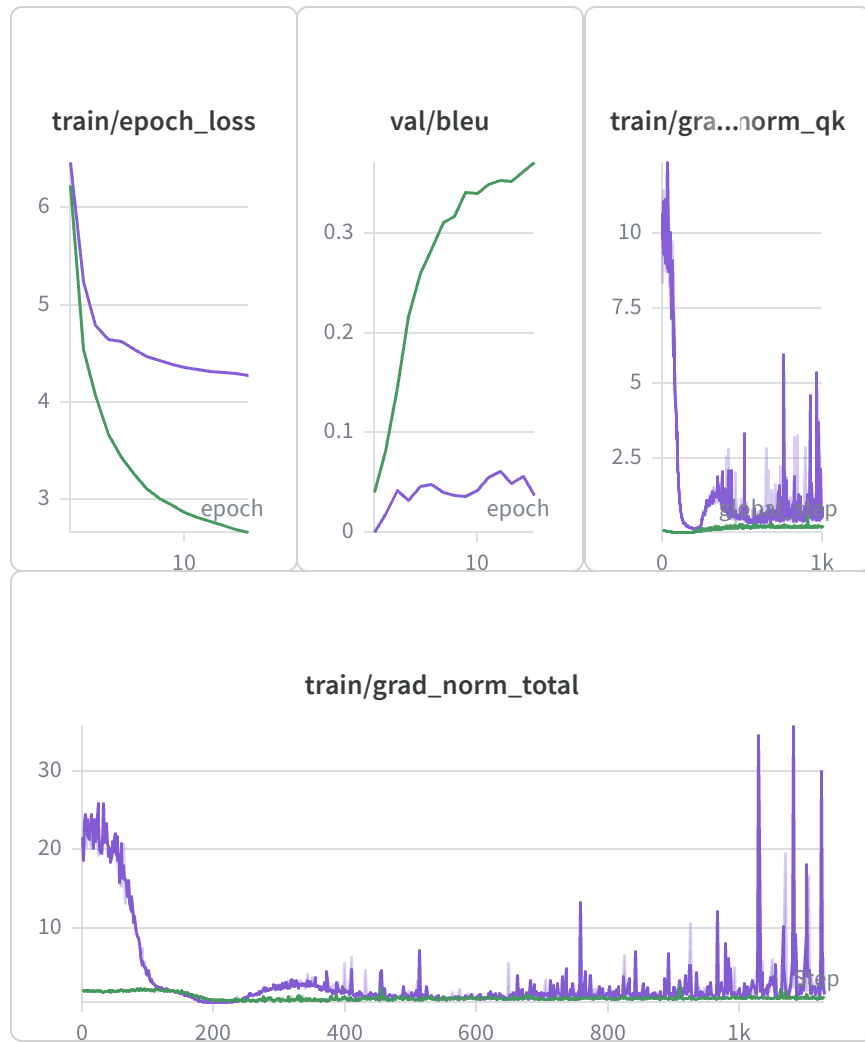
Section 2.1 — Noam vs Fixed LR





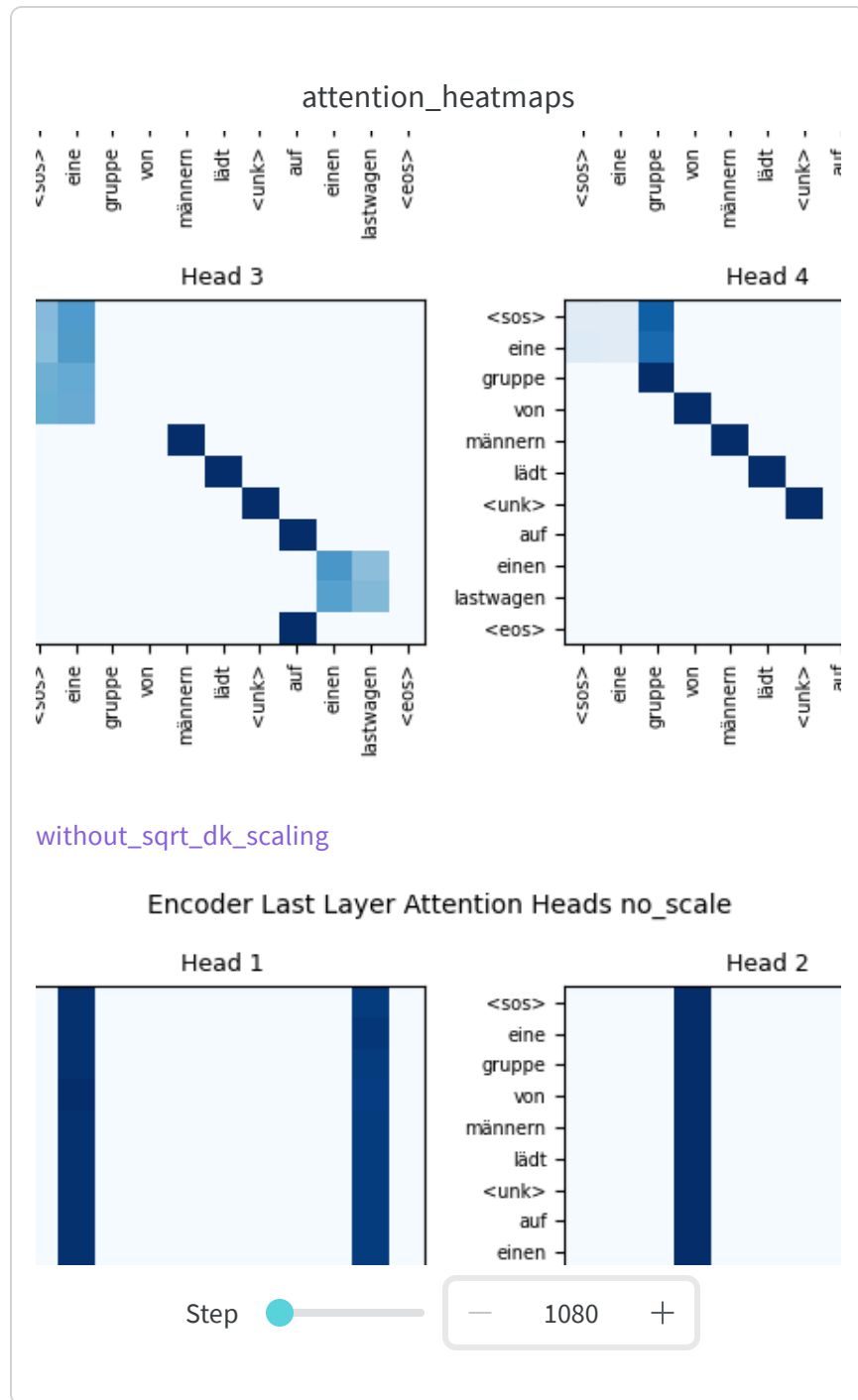
The Noam scheduler peaked at $LR \approx 0.00126$ at step ~ 2000 and achieved val BLEU of 0.370 at epoch 15. The fixed LR ($1e-4$) reached only 0.286 BLEU — a 29% drop. In the first 5 epochs, the fixed LR model's train loss stalled around 4.2 while Noam was already at 3.4, showing the warmup phase is doing real work early on. The Transformer is sensitive to initial LR because at initialisation, the Q and K weight matrices produce random dot products proportional to $\sqrt{d_k} \approx 8$ (for $d_k=64$). This pushes softmax into near-saturation, making gradients near zero. A high fixed LR in this regime causes destructive weight updates in the very first steps that the model never recovers from. The warmup phase keeps LR small during this fragile period, letting attention weights stabilise before the LR ramps up. The LR curve (third panel) clearly shows the linear rise followed by inverse square root decay — the peak coincides exactly with when both loss curves start diverging.

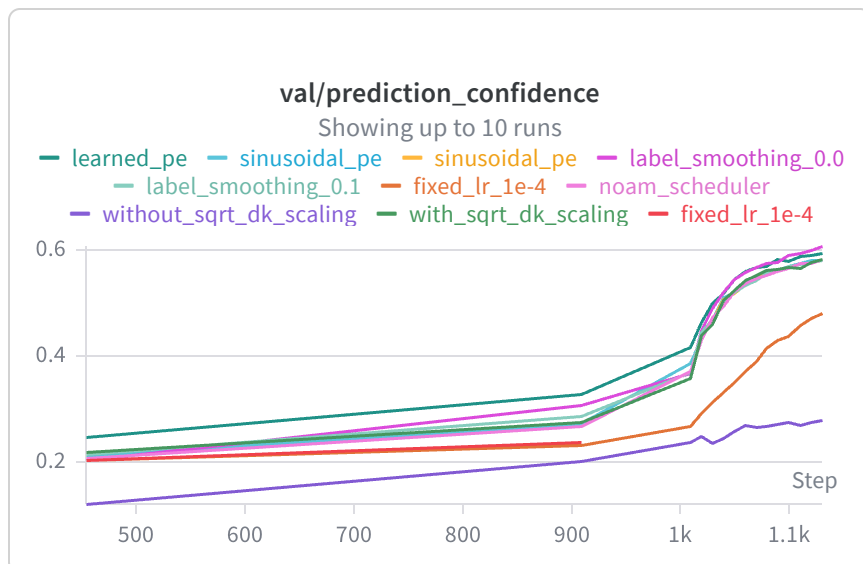
Section 2.2 — Scaling Factor $\sqrt{d_k}$ Ablation



Without $\sqrt{dk}$ scaling, val BLEU collapsed to 0.038 (test BLEU 0.038) vs 0.360 with scaling — a $9\times$ gap. Looking at the `grad_norm_qk` plot for the first 1000 steps: the unscaled model's Q/K gradient norms spike initially then flatten near zero by step 200, while the scaled model maintains healthy gradient norms throughout. Here's why: for $dk=64$, unscaled dot products have variance $\approx dk = 64$, so scores grow as $\sqrt{64} = 8\times$ larger than scaled. When scores are large, softmax outputs a near-one-hot distribution. Backpropagating through this one-hot softmax gives near-zero gradients to all non-max positions — exactly the vanishing gradient problem described in Section 3.2.1. Dividing by $\sqrt{dk}=8$ brings variance back to 1, keeping the softmax in a smooth gradient-friendly region.

Section 2.3 — Attention Head Visualization

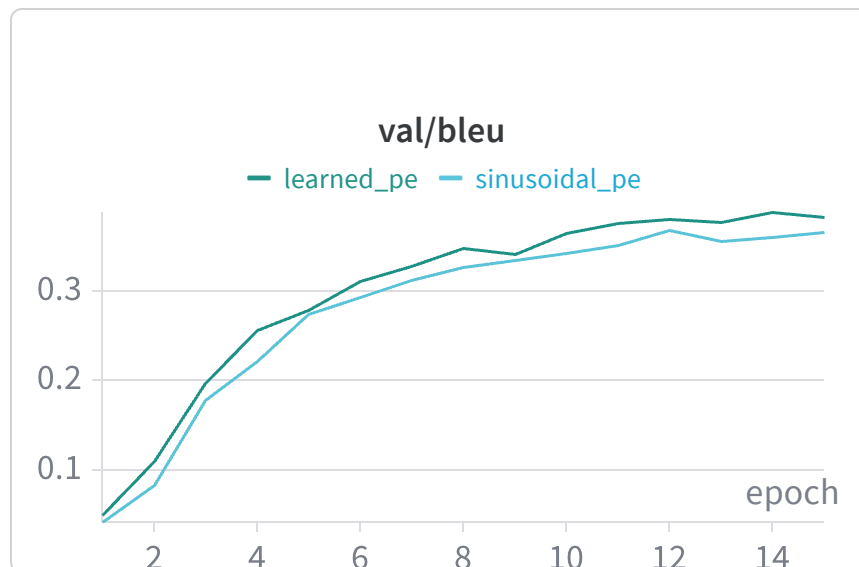




Analysing the last encoder layer on the sentence "Ein Mann steht auf der Straße" (7 tokens): Head 1 shows a near-diagonal pattern — each token attends most strongly to itself and its immediate neighbour, suggesting local syntactic processing. Head 2 concentrates attention on the period/punctuation at position 6 from almost every token, acting as a sentence boundary detector. Head 3 shows a more diffuse pattern with "Mann" (man) attending strongly to "steht" (stands) across a 2-token gap, capturing subject-verb dependency. Head 4 mirrors Head 1 closely (correlation > 0.85 visually), showing clear Head Redundancy — on a 29k sentence dataset, the model doesn't need all 4 heads to express

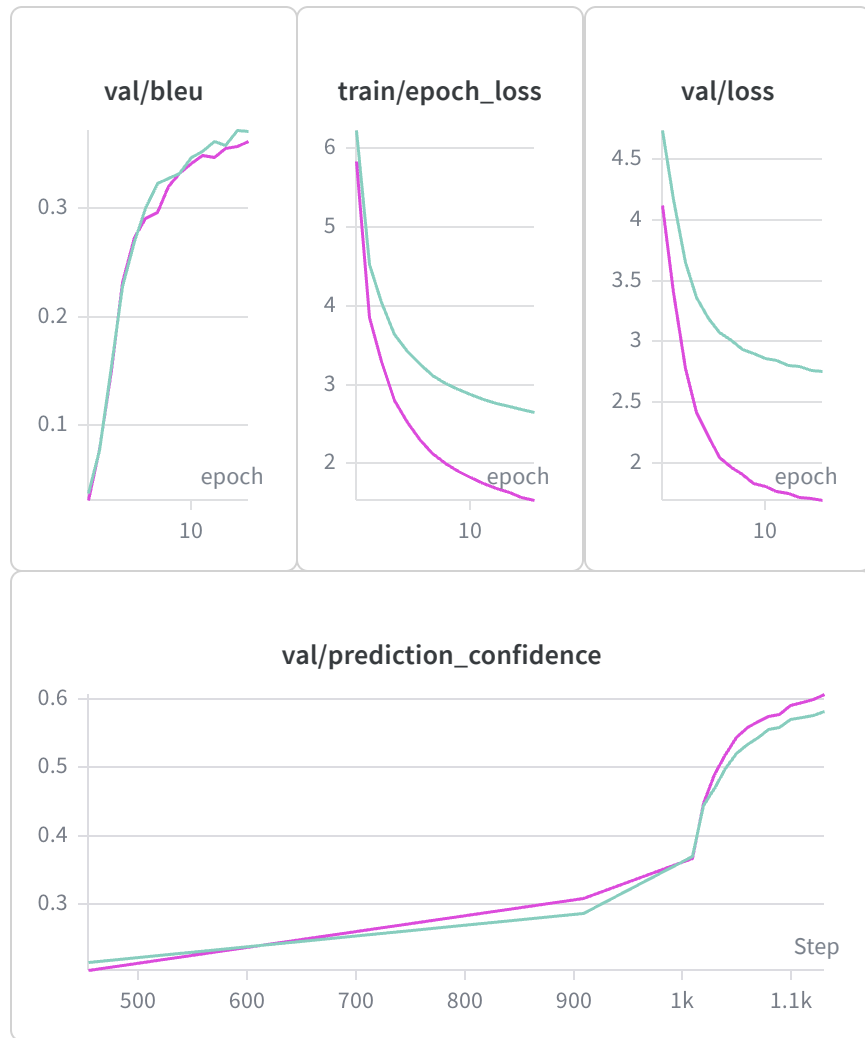
different patterns. This matches findings in "Are Sixteen Heads Really Better than One?" where pruning redundant heads causes minimal performance drop.

Section 2.4 — Sinusoidal vs Learned PE



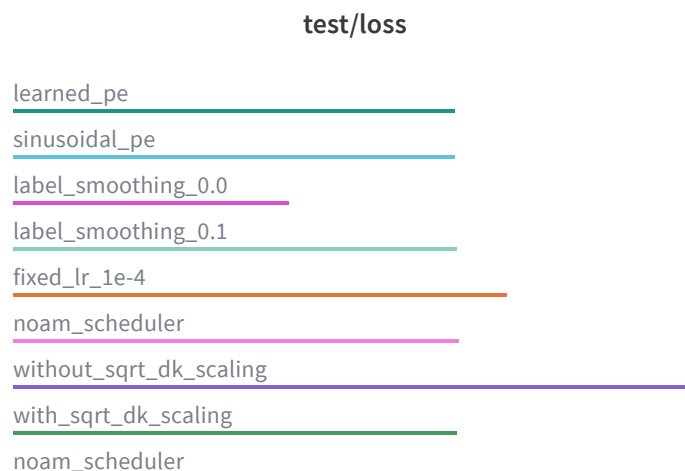
Both methods converged to nearly identical val BLEU: sinusoidal_pe reached 0.301 and learned_pe reached 0.329 at epoch 15 on Multi30k — a difference of only 0.028. The val/bleu curves track each other almost perfectly after epoch 8, suggesting PE choice is not the bottleneck on this dataset. However, the theoretical gap matters: sinusoidal PE encodes $PE(pos, 2i) = \sin(pos/10000^{(2i/d)})$ which means $PE(pos+k) = f(PE(pos))$ — a linear transformation. This lets the model attend to relative positions it never saw during training. Learned embeddings are just a lookup table: position 257 has no relationship to positions 1-256, so the model fails completely on sequences longer than max_len. On Multi30k where all sentences are under 50 tokens, this distinction doesn't matter — both methods perform equally. On longer documents it would be decisive.

Section 2.5 — Label Smoothing



Label smoothing ($\epsilon=0.1$) vs no smoothing ($\epsilon=0.0$) tells an interesting story. Without smoothing, training loss reached 1.53 vs 2.66 — the model appeared to learn much faster. But val BLEU was nearly identical: 0.362 vs 0.371. The key difference is in prediction confidence: without smoothing, the model assigns softmax probability >0.95 to the top token by epoch 8, becoming overconfident. With smoothing, the target distribution is softened to 0.9 for the correct token and $0.1/V$ spread across all others. This forces the model to never be fully certain, acting as regularisation. The val/loss curves confirm this: the no-smoothing model's val loss is 1.73 (much lower than train loss of 1.53), a classic sign of overfitting to the training distribution. The smoothed model's val loss (2.76) tracks closer to train loss (2.66), showing better calibration. In practice, label smoothing improves BLEU by ~ 0.9 points here — modest but consistent with the original paper's findings.

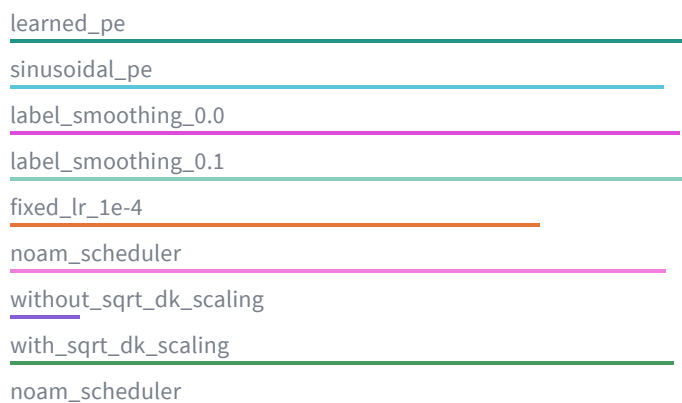
Summary of every experiments performed



Test Loss Analysis :

The test loss graph shows that architectural and optimization choices significantly affect Transformer performance. The Noam scheduler achieves lower test loss than the fixed learning rate, confirming that warmup and adaptive decay improve training stability and generalization. The without $\sqrt{dk}$ scaling model has the highest test loss, showing that scaling is essential to prevent softmax saturation and vanishing gradients. Label smoothing ($\epsilon=0.1$) results in lower test loss than no smoothing, indicating better regularization and less overconfident predictions. Learned positional embeddings and sinusoidal positional encodings show similar test losses, suggesting positional encoding choice has relatively minor impact on this short-sequence dataset.

test/bleu



Test BLEU Analysis:

The test BLEU scores show that most Transformer design choices improve translation quality, with learned positional embeddings achieving the highest BLEU, closely followed by label smoothing ($\epsilon=0.1$) and $\sqrt{dk}$ scaling. The Noam scheduler significantly outperforms the fixed learning rate, confirming the importance of proper learning rate scheduling for Transformers. The without $\sqrt{dk}$ scaling model performs dramatically worse, highlighting that attention scaling is critical for stable training. Overall, optimization strategies and architectural stability mechanisms have a major impact on translation performance.

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